Design and Evaluation of a Soft Exoskeleton Glove with Hybrid Jamming for Hand Rehabilitation

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PGT - 10

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Background

Hand weakness resulting from neurological and musculoskeletal conditions can significantly impair daily function and quality of life. Effective rehabilitation often requires repetitive, progressive, and task-specific movement training with adaptable assistance (Reinkensmeyer et al., 2004). Soft robotic exoskeleton gloves have emerged as promising rehabilitation devices due to their flexibility, safety, and ability to support natural hand movement (Rus and Tolley, 2015). Jamming, shown below, is a technique used in soft robotics to provide controllable and reversible stiffness, allowing systems to remain flexible during movement while becoming rigid when additional support or stability is required.

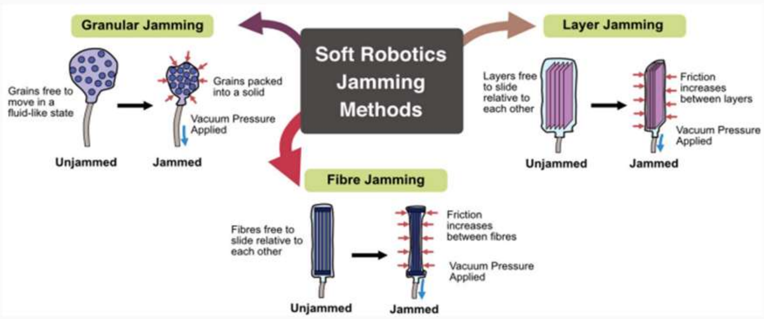
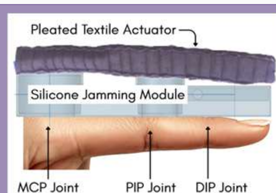
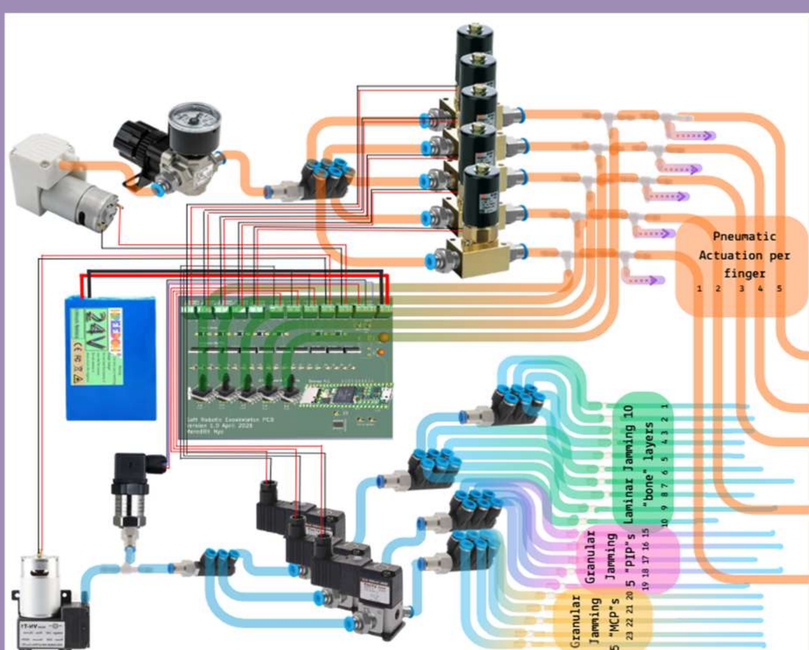


Figure adapted from: Liow L, Howard D. Jellyphant: A Soft, Elephant Trunk-Inspired Robotic Arm That Can Grab Objects. Frontiers for Young Minds. 2024.

The finger design has layered nylon fabric and mylar film for laminar jamming and EPS beads for granular jamming with oval silicone sections at the finger joints to improve fit and alignment targeting the higher load bearing joints as seen in the figure on the right.
A closed loop system will control the air pump-fabric actuator system, while the vacuum pump-jamming system will remain open loop within 8 different jamming configurations.



Finger-level design of exoskeleton glove



A representation of the full glove design including battery, custom PCB with ground and power wirings, and tubing configurations where light blue tubes represent vacuum pressure and orange tubes represent air pressure.

Aims

Aim:
To design and develop a bio-inspired soft exoskeleton glove incorporating hybrid jamming (granular and laminar) to provide adjustable, graduated support for hand rehabilitation.

Research Questions:

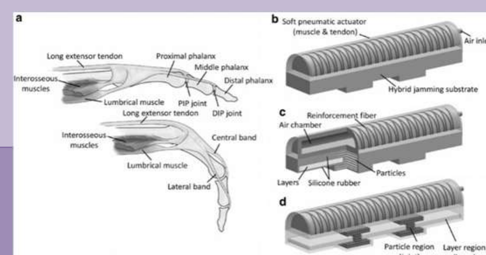
1. Can a hybrid jamming structure provide controllable and reversible stiffness and ROM suitable for assisting hand rehabilitation movements?

2. How does the design meet hand rehabilitation requirements as determined from the literature?

3. How does the design compare quantitatively to other designs in the research and on the market?

Design Rationale

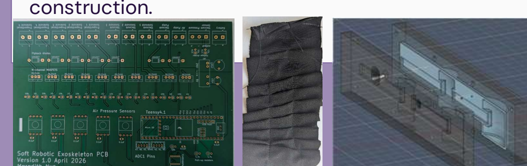
Hand rehabilitation requires repetitive, task-specific movement with adjustable assistance, sufficient range of motion, and patient-specific adaptability. Devices must also be lightweight and safe to minimise user fatigue and reduce risk during prolonged use. The current design is also heavily inspired by pleated textile actuators and the bioinspired design shown below adopted from (Yang et al., 2019).



Proposed Methodology

Testing will involve measuring force output and stiffness modulation with and without jamming, alongside ROM, actuation speed during finger flexion and extension, and power consumption. The goal is to provide a comprehensive look at the feasibility of the final design for hand rehabilitation.

Manufacturing will entail soldering custom PCB components, sewing, resin printing silicone molds, silicone pouring, and pneumatic pipe system construction.



From left to right:
Picture of the custom PCB board, hand-sewn fabric actuator mock-up, Fusion 360 model for the small half of the silicone finger's resin mold

PRESENTER:

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Team	STARS Research Group
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Reinkensmeyer, D.J. et al. (2004) Robotics in rehabilitation: A review of upper limb motor recovery after stroke.

Rus, D. and Tolley, M.T. (2015) Design, fabrication and control of soft robots.

Yang, Y., Zhang, Y., Kan, Z., Zeng, J. and Wang, M.Y. (2019) Hybrid jamming for bioinspired soft robotic fingers. Soft Robotics. doi:10.1089/soro.2018.0101

Liow, L. and Howard, D. (2024) Jellyphant: A soft, elephant trunk-inspired robotic arm that can grab objects. Frontiers for Young Minds, 12:1341887. doi:10.3389/frym.2024.1341887